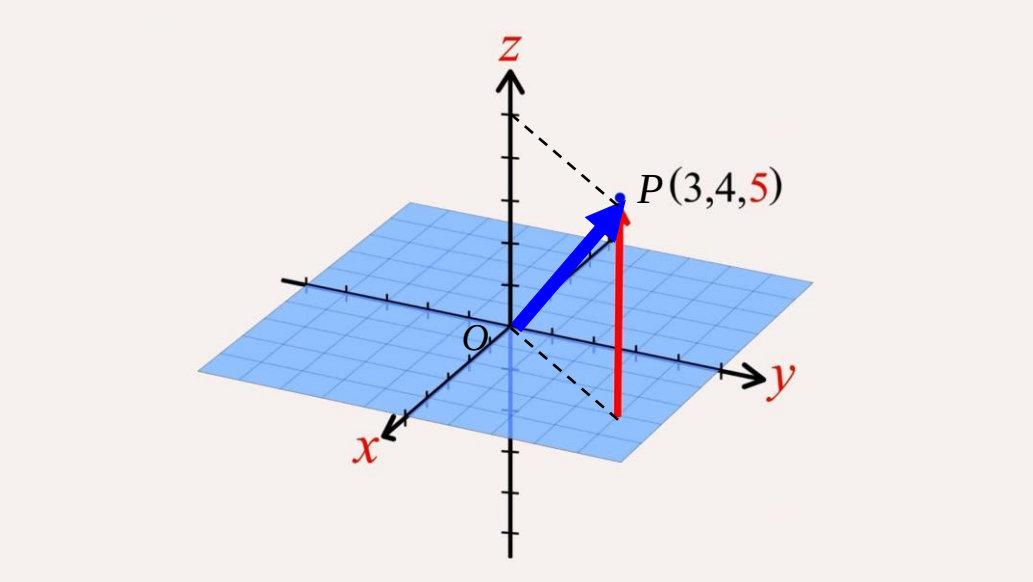
Intro. to 3-D vectors (立體向量)

In this notes, it will cover the basic concepts of 3-D vectors (i.e. vectors in $\mathbf{R}^3$).

3-Dimensional coordinates system

In 3-D space, a point can be represented as an ordered triple of real numbers (x, y, z); where x is the x-coordinate, y is the y-coordinate and z is the z-coordinate.



e.g. In the figure, the coordinates of P are (3, 4, 5)

In the 3-dimensional coordinates system, there are three **important unit vectors**:

- $\vec{i}$ represents the unit vector in the **+ve direction of *x*-axis**
- $\vec{j}$ represents the unit vector in the **+ve direction of *y*-axis**
- $\vec{k}$ represents the unit vector in the **+ve direction of *z*-axis**

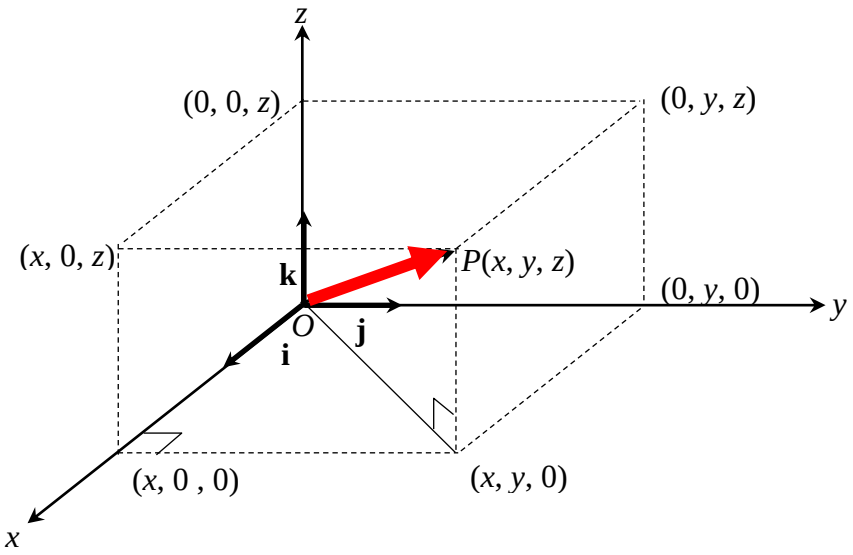
Then any point in 3-D space can be represented by a position vector,

e.g. In the figure, $\overrightarrow{OP} = 3\vec{i} + 4\vec{j} + 5\vec{k}$ and $\left|\overrightarrow{OP}\right| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50}$

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Vectors in 3-D coordinates system

Let $P\ (x\ ,\ y\ ,\ z)$ be any point in the 3-D space.



Then the position vector $\overrightarrow{OP}$ can be uniquely expressed as a linear combination of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

i.e. $\overrightarrow{OP} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ and $|\overrightarrow{OP}| = \sqrt{x^2 + y^2 + z^2}$.

Also,

if $A = (x_1\ ,\ y_1\ ,\ z_1)$ and $B = (x_2\ ,\ y_2\ ,\ z_2)$ be any two points in the 3-D space,

then $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (x_2 - x_1)\,\mathbf{i} + (y_2 - y_1)\,\mathbf{j} + (z_2 - z_1)\,\mathbf{k}$

and $|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.

e.g. If $A = (3, -4, 8)$ and $B = (5, -1, 2)$, then

$$\begin{aligned}\overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= (5 - 3)\,\mathbf{i} + [(-1) - (-4)]\,\mathbf{j} + (2 - 8)\,\mathbf{k} \\ &= 2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}\end{aligned}$$

and

$$|\overrightarrow{AB}| = \sqrt{(2)^2 + (3)^2 + (-6)^2} = 7$$

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Vectors in 3-D coordinates system

[GCE AL OCR June 2016 C4 Q.8 Modified]

Given that the position vectors of A and P are $3\mathbf{i} + 5\mathbf{j}$ and $\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$ respectively.

Find the exact value of the distance AP.

Give your answer in the form of $k\sqrt{2}$, where k is a constant. ***(2 marks)***

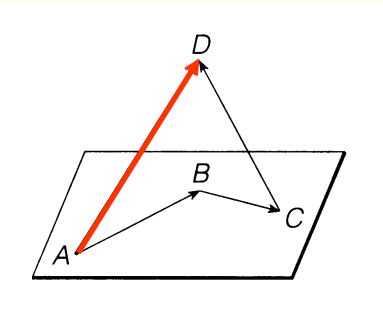
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Addition of vectors

Addition of vectors

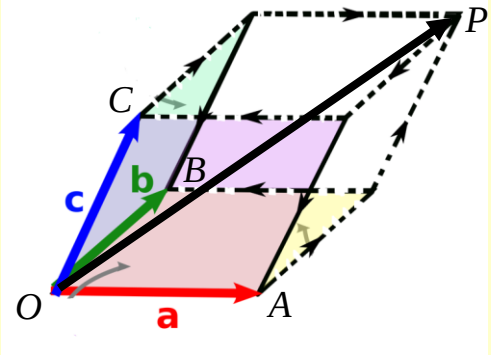
Addition of two 3-D vectors can be illustrated by Triangle law or Parallelepiped law:

1. Triangle law



$$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}=\overrightarrow{AD}$$

2. Parallelepiped (平行六面體) law



$$\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}=\overrightarrow{OP}$$

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Addition of vectors

[GCE AL OCR June 2005 C4 Q.5(a)]

ABCD is a parallelogram. The position vector of *A*, *B* and *C* are given respectively by

$\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$ $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j}$ $\mathbf{c} = \mathbf{i} - \mathbf{j} - 2\mathbf{k}$

Find the position vector of *D*. ***(3 marks)***

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Parallel vectors

Parallel vectors

Suppose there are non-zero scalars x_1, x_2, y_1, y_2, z_1 and z_2 such that

$\vec{u} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$ and $\vec{v} = x_2\vec{i} + y_2\vec{j} + z_2\vec{k}$

then

(a) $\vec{u}$ and $\vec{v}$ are **parallel** (or **collinear**) if and only if $\frac{x_1}{x_2} = \frac{y_1}{y_2} = \frac{z_1}{z_2}$

(b) $\vec{u}$ and $\vec{v}$ are **equal** if and only if $x_1 = x_2$ and $y_1 = y_2$ and $z_1 = z_2$

e.g.

- $2\vec{i} + 3\vec{j} - 4\vec{j}$ and $-4\vec{i} - 6\vec{j} + 8\vec{k}$ are parallel;
- $2\vec{i} + 3\vec{j}$ and $2\vec{i} + 3\vec{j} - 4\vec{k}$ are not parallel

Collinear points in 3-D space

Three point A, B and C are **collinear** if and only if

$\overrightarrow{AB} = k\overrightarrow{BC}$ (or $\overrightarrow{AB} = k\overrightarrow{AC}$ or $\overrightarrow{BC} = k\overrightarrow{AC}$) where k is non-zero constant

e.g. Given three points $A(-1, 6, 5)$, $B(2, 3, 8)$ and $C(3, 2, 9)$. Since

$$\begin{aligned}\overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA}\\ &= [(2-(-1)]\,\mathbf{i} + (3-6)\,\mathbf{j} + (8-5)\,\mathbf{k}\\ &= 3\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}\end{aligned}$$

$$\begin{aligned}\overrightarrow{BC} &= \overrightarrow{OC} - \overrightarrow{OB}\\ &= (3-2)\,\mathbf{i} + (2-3)\,\mathbf{j} + (9-8)\,\mathbf{k}\\ &= \mathbf{i} - \mathbf{j} + \mathbf{k}\end{aligned}$$

$\because$ $\overrightarrow{AB} = 3\overrightarrow{BC}$

$\therefore$ A, B and C are collinear.

Parallel vectors



[GCE AL OCR Jan 2010 C4 Q.2(a)]

Points A, B and C have position vectors $-5\mathbf{i} - 10\mathbf{j} + 12\mathbf{k}$, $\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $3\mathbf{i} + 6\mathbf{j} + p\mathbf{k}$ respectively, where p is a constant.

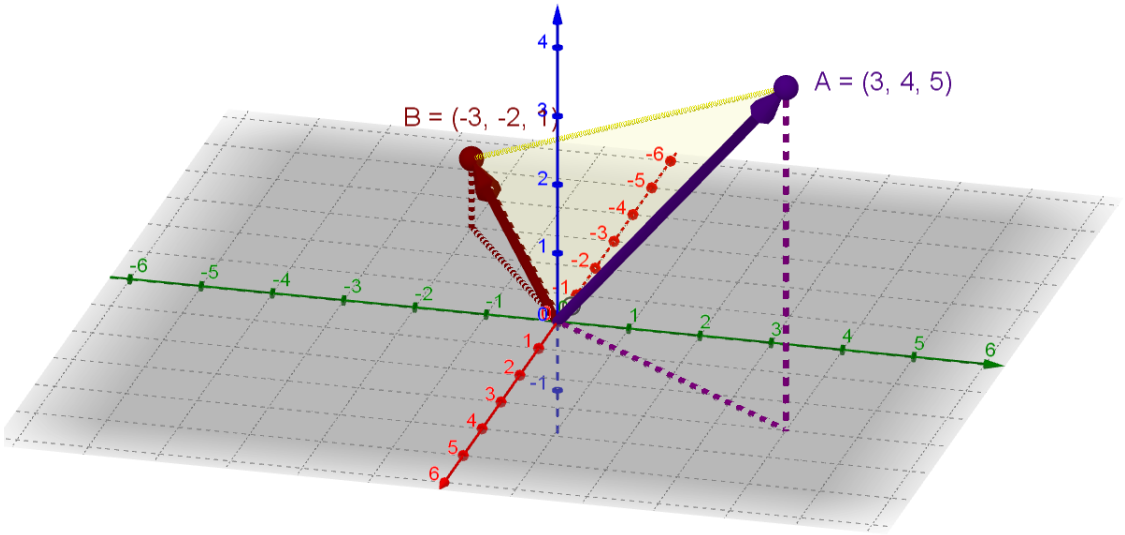
Given that ABC is a straight line, find the value of p. ***(2 marks)***

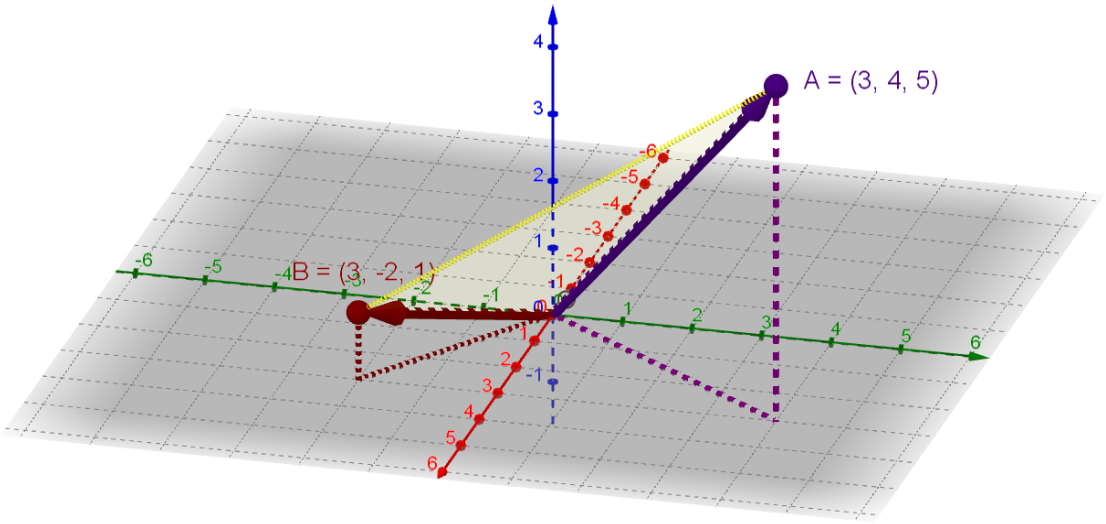
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Coplanar vectors

Vectors are **coplanar (共面)** if they lie on the same plane.

Note that any two position vectors / free vectors in $\mathbf{R}^3$ must be coplanar.





For any three vectors **a**, **b** and **c** in $\mathbf{R}^3$, we can use **Scalar Product and Vector Product** to check whether they are coplanar. It will be discussed later.

Section Formula

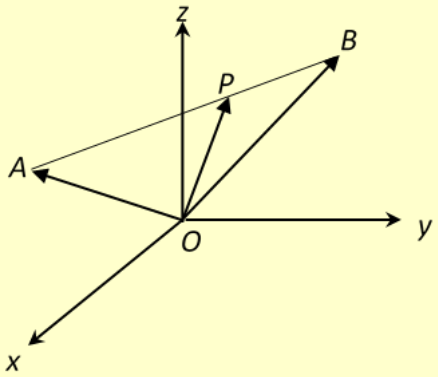
Section formula in vectors

For any points A and B with another point O,

(a) if $\boldsymbol{AP : PB = m : n}$,

$$\overrightarrow{OP} = \frac{n\overrightarrow{OA} + m\overrightarrow{OB}}{m+n}$$

Note that $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OP}$ are coplanar.



(b) if M is the **mid-point** of AB,

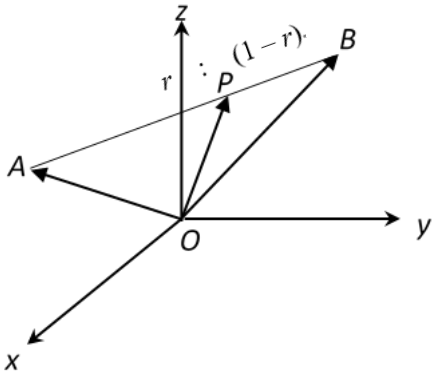
$$\overrightarrow{OM} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2}$$

Note that $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OM}$ are coplanar.

Note: If we let $\boldsymbol{AP : PB = r : (1 - r)}$, then we have

$$\mathbf{p} = (1 - r)\mathbf{a} + r\mathbf{b}$$

where **p**, **a** and **b** are position vectors of P, A and B respectively. Furthermore,



(i) P lies between A and B for $0 < r < 1$

(ii) P lies on BA produced for $r < 0$

(iii) P lies on AB produced for $r > 1$

Moreover for any point P lies on the straight line passing through A and B, its position vector **p** can be represented by

$$\mathbf{p} = \overrightarrow{OA} + \lambda\overrightarrow{AB} \qquad \text{where } \lambda \text{ is real values}$$

and it is well known as **vector equation** (i.e. equation of straight line in $\mathbf{R}^3$)

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Section Formula

[GCE AL WJEC June 2013 C4 Q.9(a)(b)]

The position vectors of the points A and B are given by

$$\mathbf{a} = -\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$$

$$\mathrm{b} = 7\mathbf{i} - \mathbf{j} + 5\mathbf{k}$$

respectively.

(a) Write down the vector AB. ***(1 mark)***

(b) The point C lies on the line AB such that $AC : CB = 3 : 1$.

Find the position vector of C. ***(2 marks)***

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More Questions from Past Papers

[GCE AL Edexcel Jan 2007 C4 Q.7(a)(b)(c)]

The point A has position vector $\mathbf{a} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and the point B has position vector $\mathbf{b} = \mathbf{i} + \mathbf{j} - 4\mathbf{k}$, relative to an origin O.

(a) Find the position vector of the point C, with position vector $\mathbf{c}$, given by

$$\mathbf{c} = \mathbf{a} + \mathbf{b}.$$

(1)

(b) Show that $OACB$ is a rectangle, and find its exact area.

(6)

The diagonals of the rectangle, AB and OC, meet at the point D.

(c) Write down the position vector of the point D.

(1)

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HKDSE Questions (level 3)

1. ***[M2 2022 Q.12]***

Consider $\triangle ABC$. Denote the origin by O.

(a) Let D be a point lying on BC such that AD is the angle bisector of $\angle BAC$.
Define $BC = a$, $AC = b$ and $AB = c$.

(i) Using the fact that $BD : DC = c : b$, prove that

$$\overrightarrow{AD} = -\overrightarrow{OA} + \frac{b}{b+c}\overrightarrow{OB} + \frac{c}{b+c}\overrightarrow{OC}.$$

(ii) Let E be a point lying on AC such that BE is angle bisector of $\angle ABC$.

Define $\overrightarrow{OJ} = \frac{a}{a+b+c}\overrightarrow{OA} + \frac{b}{a+b+c}\overrightarrow{OB} + \frac{c}{a+b+c}\overrightarrow{OC}$.

Prove that J lies on AD. Hence, deduce that AD and BE interesct at J.

(7 marks)

(b) Suppose that $\overrightarrow{OA} = 35\mathbf{i} + 9\mathbf{k} + \mathbf{j}$, $\overrightarrow{OB} = 40\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ and $\overrightarrow{OC} = -3\mathbf{j} + \mathbf{k}$. Let I be the incenter of $\triangle ABC$. Find $\overrightarrow{OI}$. ***(2 marks)***

HKDSE Questions (level 3)

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